Huawei Access Node (JAY-ACC-68) BGP Route Churn and OSPF Adjacency Loss Troubleshooting SOP (ALARM_4384)

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4384

Description

This Standard Operating Procedure is designed to address a critical failure scenario where the Huawei Access Node (JAY-ACC-68) at the Jayanagar Metro Site experiences a Routing Protocol Flap, characterized by a high rate of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This phenomenon is primarily attributed to the intricate interplay between the Border Gateway Protocol (BGP) and Open Shortest Path First (OSPF) routing protocols, which are responsible for establishing and maintaining routing table entries between routers within the network. The BGP route churn occurs when the BGP protocol detects a change in the network topology, leading to a re-advertisement of routes, which in turn triggers a cascade of route updates throughout the network. Concurrently, the OSPF adjacency loss indicates a disruption in the OSPF protocol's ability to establish and maintain neighbor relationships between routers, resulting in the loss of routing information. The ISIS LSP flooding, on the other hand, signifies an excessive generation of link-state packets (LSPs), which can overwhelm the network, leading to performance degradation and potential network instability. The root cause of this issue may lie in misconfigured BGP or OSPF settings, hardware malfunctions, or software glitches.

Pre-requisite Checks

- Check: Verify the power status of the Huawei Access Node (JAY-ACC-68) at the Jayanagar Metro Site., Action: Dispatch a remote hands team to check the power status of the node and ensure that it is properly connected.

- Check: Inspect the network topology and verify the connectivity between the Huawei Access Node (JAY-ACC-68) and its neighboring routers., Action: Use network management tools to visualize the network topology and ensure that there are no physical or logical link issues.

- Check: Verify the redundant link status between the Huawei Access Node (JAY-ACC-68) and its neighboring routers., Action: Check the configuration of the redundant links and ensure that they are properly set up and functioning as intended.

Potential Root Causes

Cause	Probability
Misconfigured BGP settings	High
Hardware malfunction in the Huawei Access Node (JAY-ACC-68)	Medium
Software glitch in the Huawei Access Node (JAY-ACC-68)	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: BGP routing-table information, Description: This command displays the BGP routing table, allowing us to identify any issues with route churn or instability.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays the OSPF neighbor table, enabling us to identify any issues with adjacency loss or routing information loss.

Command: display isis lsp, Expected Output: ISIS LSP information, Description: This command displays the ISIS LSP information, allowing us to identify any issues with LSP flooding or excessive LSP generation.
```

Resolution Steps

Step 1: Step: Verify and correct BGP settings, Command: display bgp configuration, Description: This command displays the BGP configuration, allowing us to identify any misconfigured settings that may be contributing to the issue.

Step 2: Step: Verify and correct OSPF settings, Command: display ospf configuration, Description: This command displays the OSPF configuration, enabling us to identify any misconfigured settings that may be contributing to the issue.

Step 3: Step: Perform a hardware reset on the Huawei Access Node (JAY-ACC-68), Command: reset system, Description: This command performs a hardware reset on the node, which may resolve any hardware-related issues.

Post-Check Verification

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Command: display bgp routing-table, Expected Output: BGP routing-table information, Description: This command displays the BGP routing table, allowing us to verify that the issue has been resolved.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays the OSPF neighbor table, enabling us to verify that the issue has been resolved.

Command: display isis lsp, Expected Output: ISIS LSP information, Description: This command displays the ISIS LSP information, allowing us to verify that the issue has been resolved.
```

Roll-back Procedures

- Step: Restore the previous BGP settings, Command: undo bgp configuration, Description: This command restores the previous BGP settings, which may resolve the issue.

- Step: Restore the previous OSPF settings, Command: undo ospf configuration, Description: This command restores the previous OSPF settings, which may resolve the issue.

- Step: Perform a hardware reset on the Huawei Access Node (JAY-ACC-68), Command: reset system, Description: This command performs a hardware reset on the node, which may resolve any hardware-related issues.